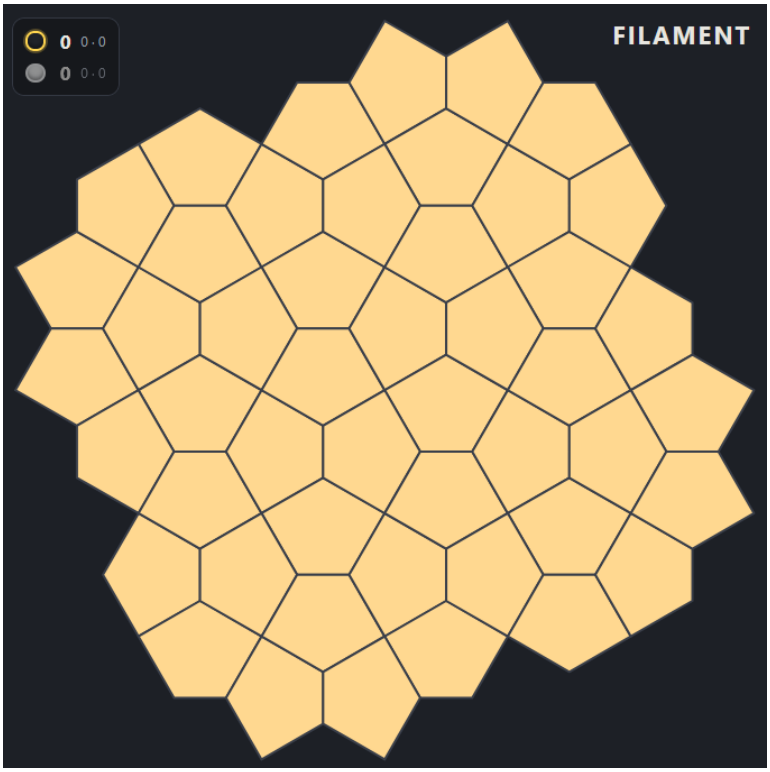
Filament

2026, Brian E. Svoboda, https://boardgamegeek.com/thread/3775896

The game is played on the following pentagonal tiling:



Groups. Two of your stones are in the same group if you can trace a path between them that steps only from neighbor to neighbor through your own stones. A stone with no neighbors of its color is a group by itself.

Setup. Play on a patch of the Cairo pentagonal tiling where every cell has either 3 or 5 neighbors. Two cells are neighbors if they share an edge; cells that touch only at a corner don't count. The board starts empty, and Black moves first.

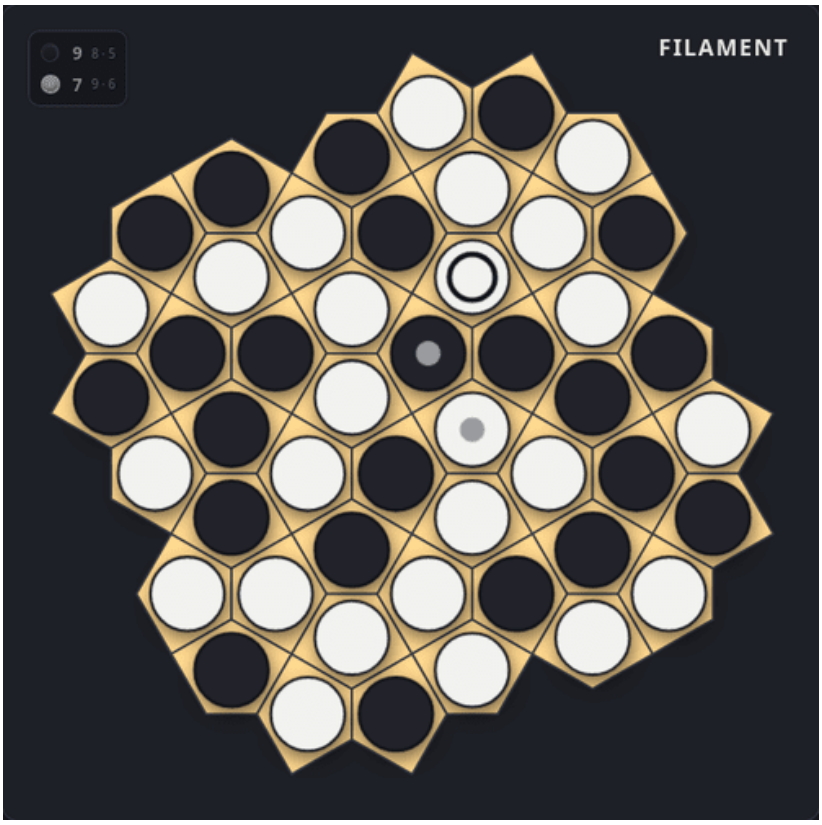
Place. On your turn place a stone of your color on an empty cell.

Fill. After a placement, determine if any neighboring empty cells have a majority where more than half of all its neighboring cells hold stones of one color: at least 3 of 5 for inner cells, or at least 2 of 3 for edge cells. Fill each such cell with a stone of the opposite color of the majority. Fills can create new majorities. Continue filling until none remain. Then your turn ends.

Game end. The game ends when the board is full. The player with fewer groups wins.

Ties. If you tie on group count, list your group sizes from largest to smallest. Compare your list against your opponent's one position at a time: first the largest

group, then the second-largest, and so on. At the first position where the sizes differ, the player with the smaller group wins. If every size matches, the game is a draw.



An example game won by White. Last move shown by the open circle and filled stones shown with gray filled circles.

Author notes: The most obvious unique aspect of an isohedral pentagonal tiling compared to hex or square is that cells have an odd number of neighbors, i.e., five. When playing with two colors of stones, this means every cell's neighborhood will have a clear majority in favor of one color at 3/5, 4/5, or 5/5. Not all board shapes have all odd degree though, most will include some edge cells with four neighbors. But I did find two patches where all cells have either 3 or 5 neighbors. These have 48 and 160 total cell count, roughly equivalent to 7x7 and 13x13 on a square grid, respectively. So the design process mainly involved seeing how to turn this into a viable mechanism and pair it with a complementary win condition.

The most interesting combination I found involved filling such local majorities with the **opposite** color. This has an "anti-blobbing" effect that promotes fragmentation, and when paired with a "fewest groups wins" goal, leads to nice strategic tension. The "one big central blob" strategy fails because your opponent can force isolated stones of yours onto the edge of the board. However, the fewest groups goal isn't decisive on its own, so ties are broken by whoever has the smaller largest group (cascading, with draws possible if all groups are equal). This means

that in close games extending or connecting your large groups may not to be the right thing to do, because in effect the tie-break favors the player with more uniform group sizes.

The filling mechanism provides a nice pacing between "small" and "large" turns. Most turns are single placements but there can be some dramatic cascades near the end. The unusual board and fills both make it on the opaque side, but I think that's okay because it makes a pure placement game more interesting on a small board.

Some caveats: it is broken by mirror play as-is, so needs a two-stone pie for competitive play or a house rule not to mirror on the second move for casual play. Scorekeeping is tedious but sort of goes with the territory here. For OTB I'd recommend a score track for the group count. The digital implementation displays group counts as well as the two largest group sizes per player. The 48-cell board goes pretty quick because of the fills with a typical game length of ~15 turns. The 160-cell board is probably closer to ~45-60 turns.